Tejas Nehete

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Education

Vellore Institute of Technology **2023 – 2027**
B.Tech in Computer Science and Engineering *8.95/10 CGPA*

Experience

Indian Institute of Technology Indore **May 2025 – June 2025**
Summer Research Internship

- Contributed to an **ICU event tracking and data collection research project**, focusing on computer vision, dataset preparation, and machine learning components for detecting ICU devices and events.
- Trained and fine-tuned **YOLOv8** models on sample and medical image datasets, working with preprocessing, annotation formats, model training, and evaluation.
- Explored **YOLOv7 incremental learning** for adapting object detection models to newly introduced ICU object classes while retaining previously learned knowledge.
- Annotated **500+ AIIMS medical images** covering **4 ICU object classes**: Ventilator, Monitor, Bed_no, and Face, supporting dataset preparation for model training and evaluation.
- Annotated **250+ images** for Qwen-based experiments and contributed to the preparation and annotation of a dataset containing **6,000+ images** for computer vision model development.
- Researched **Qwen + LoRA** and **Grounding DINO + SAM** for automated image annotation and improving the efficiency of large-scale computer vision data preparation.

Technical Skills

Programming Languages: C++, Java, Python, JavaScript, HTML, CSS
Frameworks & Libraries: React.js, Next.js, Node.js, Express.js, Tailwind CSS, TensorFlow, Keras, Scikit-learn, OpenCV, NumPy, Pandas
Databases & Tools: MySQL, MongoDB, Git, Docker, REST APIs
Machine Learning: YOLOv7, YOLOv8, Qwen, LoRA, Grounding DINO, SAM, Object Detection, Image Annotation, Incremental Learning

Projects

PaperTrade Simulator **Link**

- Engineered a full-stack web platform simulating stock trading with real-time market data through the **Finnhub API**.
- Architected **RESTful APIs using Node.js and Express.js** for transaction processing, portfolio management, balance updates, and trade history.
- Secured user accounts through **JWT authentication** and persisted portfolios, trades, and balances using MongoDB.
- Designed responsive **React.js dashboards** with Tailwind CSS for executing trades and monitoring portfolio performance.
- **Technologies:** React.js, Tailwind CSS, Node.js, Express.js, MongoDB, JWT, REST APIs, Finnhub API.

Notely.ai **Link**

- Developed a full-stack web application for uploading PDFs and asking context-aware questions through an AI-powered interface.
- Integrated the **Google Generative AI API** to process uploaded documents and generate context-aware responses.
- Implemented **secure authentication using Clerk** and integrated Convex for backend data management and synchronized application state.
- Crafted a responsive interface using **Next.js and Tailwind CSS** with PDF processing capabilities.
- **Technologies:** Next.js, Tailwind CSS, Google Generative AI API, Clerk, Convex, PDF.js.

ChurnScope **Link**

- Constructed a full-stack analytics dashboard for customer churn prediction and sales forecasting.
- Trained ML models using **Python and scikit-learn** to identify high-risk customers and segment them by churn probability.
- Visualized **churn trends, sales patterns, and demand forecasts** through interactive dashboards.
- Processed datasets using **Pandas and NumPy** and generated analytical visualizations with Matplotlib/Seaborn.
- **Technologies:** React.js, Python, Flask, scikit-learn, Pandas, NumPy, Matplotlib, MySQL/MongoDB.

Achievements

- **IET Book Chapter Publication** – Co-authored *Chapter 16*, "Overview of Lung Cancer Detection: A Short Survey," published in *AI, Numerical Optimization, IoT and Blockchain for Healthcare 4.0*, IET, **November 2025**.
- **Innov-era Grand Finale** – Selected among the **Top 5 Teams** for the Grand Finale of Innov-era 2025 at K.K. Wagh Institute of Engineering Education and Research, Nashik.
- **AWS Educate** – Earned **4 technical badges** in **Databases, Networking, Serverless, and Storage**.

Certifications

- The Bits and Bytes of Computer Networking
- NPTEL – Marketing Analytics (**Gold + Elite, Top 5%**)
- NPTEL – Introduction to Machine Learning (**Elite**)